

Replication: Negative Effects of Social Triggers on User Security and Privacy Behaviors

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Abstract—The social influence of peers, family, and online communities significantly impacts user security and privacy (S&P) decisions. While researchers have focused on the beneficial aspects of this influence, the potential for detrimental social effects, which may encourage users to engage in risky behaviors, remains a critical area of study. This paper presents a replication study of the work by Moore et al. (2024) [1], which investigated the negative effects of social triggers on S&P behaviors. We conduct this replication within the unique cultural and social context of Saudi Arabia, a collectivist society, to understand how cultural dimensions affect the prevalence and nature of these social triggers. Our findings confirm that social triggers are a significant factor in risky S&P behaviors in the Saudi context. We found that Family and Friends were the most frequent source of these triggers, highlighting the strong influence of in-group members. Furthermore, we observed a “negative chain” effect, where socially triggered behaviors were more likely to be shared with others. Our results suggest that more efforts are needed to reduce negative social effects, and we propose specific approaches for designing culturally-aware security interventions.

1. Introduction

Human beings are intrinsically social, and their security and privacy (S&P) decisions are significantly influenced by those around them. Social influence in this context appears in multiple forms, including direct advice from family members and friends, observation of peer behavior, and conformity to group expectations [2], [3], [4], [5], [6], [7], [8]. Prior research has extensively documented the positive potential of these influences, showing that users frequently adopt secure practices when they receive beneficial advice or observe positive examples from their social environment [9], [10]. Such practices often lead to what is described as *positive social chains*, where secure behaviors propagate within a community.

However, the potential for detrimental social effects remains significantly under-explored. Negative social triggers may encourage users to engage in insecure practices, including sharing credentials, delaying updates, or reusing weak passwords. These influences can undermine security education efforts and may contribute to the normalization of risky behaviors across communities.

Moore, Mori, and Hasegawa (2024) [1] directly addressed this gap by investigating the negative impact of social triggers on S&P behaviors in a large-scale study in the United States. Their findings showed that approximately 20–50% of risky S&P behaviors were socially triggered, either through advice or through observation of others performing the behavior. Moreover, they demonstrated that these socially triggered behaviors were significantly more likely to be shared with others, forming a *negative chain* of insecure practices. They concluded by emphasizing the need for replications in other cultural contexts, as the prevalence and source of social triggers may differ across societies.

This paper presents a replication of Moore et al. (2024) conducted in Saudi Arabia, a context that differs substantially from the U.S. in terms of cultural norms, social expectations, and the role of social groups. Existing literature on cultural dimensions describes Saudi Arabia as a highly collectivist society [8], [11], where in-group members, particularly family and close friends, hold considerable influence on personal decision-making. This cultural difference suggests that the mechanisms and origins of social triggers for S&P behaviors may vary significantly from those observed in individualistic societies.

Our study aims to examine whether socially triggered risky S&P behaviors generalize to this collectivist, non-Western context, and how cultural dynamics influence the underlying mechanisms. Specifically, we address the following research questions, which mirror those of the original study [1]:

RQ1: How frequent are social triggers for risky user behaviors in Saudi Arabia?

RQ2: By whom are users triggered to engage in risky behaviors?

RQ3: What are the factors behind these social triggers?

RQ4: How frequently, and why, do users share their risky behaviors with others?

To answer these questions, we conducted structured interviews with participants in Saudi Arabia, focusing on six categories of risky S&P behaviors: password practices, account sharing, update management, internet connections, content downloads, and social media posts. Participants were asked whether they had engaged in these behaviors within the last six months, the triggers that preceded them, the

individuals associated with these triggers, and their sharing practices.

Our results show that risky online behaviors were common among participants. For example, 84.8% of participants reported password reuse and 63.6% reported delaying software updates. Family and Friends were the most frequent source of both observed behaviors and direct advice. Participants often reported that their risky behaviors were motivated by convenience or shaped by cultural and social norms. We also found evidence of a negative chain: socially triggered risky behaviors were more likely to be shared with others, often with the intention of helping or informing them. Those who did not share frequently stated that the behavior was too common or trivial to mention. Gender differences were also observed; male participants (41%) were slightly more likely than female participants (33%) to engage in risky behaviors, and females were more influenced by online influencers compared to males.

Overall, our key contributions are as follows:

- A systematic replication of Moore et al. (2024) [1] in a non-Western, collectivist cultural context, addressing the call for cross-cultural validation.
- An analysis of how cultural factors shape the prevalence and sources of social triggers, demonstrating that the strong collectivist nature of Saudi society results in a greater reliance on in-group members as sources of negative social influence.
- Practical recommendations for designing culturally-aware S&P interventions that account for the strong influence of social and cultural norms in the Middle East.

2. Related Work

Our study builds upon and extends the growing body of research investigating the social dimensions of cybersecurity, with a specific focus on the negative behavioral influences within the Saudi Arabian context. This section reviews literature on risky security behaviors, the powerful role of social influence, and the critical gap in culturally specific studies that our replication addresses.

2.1. The Original Study

In 2024, Moore et al. [1] conducted a large-scale survey study that examined the behavioral triggers that lead users in the United States to engage in risky security and privacy behaviors. Their work focused on identifying when these behaviors occurred, what or who influenced them, and whether risky behaviors were later shared with others. Moore et al. collected data from 417 participants through an online survey and investigated six common risky behaviors: connecting to public Wi-Fi, reusing passwords, downloading illegal or unofficial content, delaying software updates, sharing sensitive information online, and sharing accounts.

The authors found that risky behaviors were highly prevalent among participants, with more than 90% reporting

that they had engaged in at least one of the six behaviors. A central contribution of the study was its focus on social triggers. 20–50% of the risky behaviors were socially influenced, most often through observing the behavior of others or receiving informal advice. Friends, family members, colleagues, and even online acquaintances played a role in shaping users' decisions, indicating that social contexts can reinforce unsafe digital habits.

Moore et al. also observed that behaviors which were socially triggered were more likely to be subsequently recommended or passed on to others. This created what the authors described as negative social chains in which risky behaviors spread unintentionally across social networks. The motivations behind sharing were typically framed as helping others gain convenience or achieve a desired outcome, rather than intentionally promoting insecure practices.

Demographic variables such as age and gender showed some association with how frequently participants engaged in risky behaviors. However, these demographic factors were not strong predictors of whether a behavior was socially triggered. Instead, the type of risky behavior itself was the clearest indicator of social influence. Behaviors such as account sharing and illegal downloading were especially likely to be encouraged or observed within social circles.

Moore et al. concluded that harmful peer influence plays a significant role in shaping everyday security practices. They emphasized the importance of recognizing negative social effects when designing educational interventions and recommended focusing on the specific behaviors that propagate most easily through social chains.

2.2. Risky User Behaviors

A consistent finding in usable security literature is the persistent gap between expert-recommended practices and actual user behavior. Non-expert users frequently engage in risky practices such as password reuse, ignoring software updates, and using unsecured public Wi-Fi [12] [13]. Early work by Ion et al. [12] and Busse et al. [13] established that security measures like password managers and two-factor authentication (2FA) see low adoption among general users, often due to perceived inconvenience or lack of technical knowledge [14], [15]. This trend continues, with recent studies confirming that convenience remains a primary driver for password re-use, even among users who are aware of the associated risks [16]. Herley's seminal work [17] and Tam et al. [16] argued that users often make rational decisions to ignore security advice when the perceived cost (e.g., time, effort) outweighs the perceived risk, a finding that continues to shape the field. Furthermore, users' decisions are often complicated by misconceptions about security technologies [9], [15], [18], [19]. For instance, Kelley et al. [18] demonstrated that users often have a poor mental model of how their online actions translate to privacy risks, and Abu-Salma et al. [19] suggested that specific misconceptions limit user motivation to adopt secure tools. Our study focuses on a set of these common, high-prevalence risky behaviors—including password practices, software updates,

and account sharing—to understand the social forces that perpetuate them in a new cultural context. Specifically, we seek to understand if the social triggers identified in the original study are equally potent in driving these established risky behaviors in Saudi Arabia.

2.3. Social Effects on User Behaviors

Human decision-making is profoundly social, and security behaviors are no exception. Research has robustly demonstrated the dual potential of social influence in the S&P domain, which can be broadly categorized into positive and negative effects.

Positive Social Effects. Foundational work by Das et al. [9] systematically typified the triggers for security behaviors, identifying “social triggers”—where users interact with or observe others—as the most common, followed by proactive and forced triggers. Their study revealed that social influence affects the likelihood of adopting a security feature, with the effect being stronger for features that are more observable by peers. They also discovered a “positive chain” effect, where socially triggered secure behaviors were four times more likely to be shared with others, suggesting a mechanism for the positive propagation of secure practices. This aligns with earlier findings by Rader et al. [?] and Pfeffer et al. [5] who showed that users often learn about security through stories and advice from friends, family, and colleagues, with these informal lessons significantly impacting their subsequent security decisions.

More recent work has explored leveraging this effect through design. Krsek et al. [7] found that social suggestions, whether from experts or the public, significantly increased the adoption of privacy settings in a study of Facebook users, demonstrating the power of crowdsourced social proof in influencing S&P decisions. Similarly, Kropczynski et al. [20] examined the concept of “Tech Caregiving” within small social groups made up of friends, family members, and coworkers. Their findings showed that tech caregiving is not a fixed role. Instead, people often shift between giving and receiving help. They also observed that older adults and young adults were more frequently the ones receiving support rather than providing it.

Redmiles et al. highlighted the presence of a digital divide in how people receive security advice. Their work showed that individuals with higher technical skill, who are often more socioeconomically advantaged, were more likely to rely on guidance from the workplace. In contrast, those with lower technical skill were more likely to depend on advice from family members and friends [21].

Building on these patterns of interaction among users, prior work has proposed several collaborative and community-focused strategies to improve security and privacy behaviors [3], [4], [7], [22], [23], [24]. For instance, Das et al. demonstrated that interventions based on social proof can successfully motivate users to explore security features by showing them that their friends already use those features [3]. Krsek et al. [7] found that participants were significantly more likely to adopt recommended S&P

settings when those settings were presented as suggestions from experts or from the general public, compared to the default Facebook settings. Notably, they observed no meaningful difference between the influence of expert suggestions and suggestions attributed to other users.

Wash and Cooper [24] performed a field experiment on phishing training that incorporated social narratives. Their findings showed that factual information and direct advice are most effective when delivered by security professionals, while narrative-based stories have greater impact when they come from individuals perceived as similar to the participant.

Negative Social Effects. In contrast to the well-studied positive effects, the negative aspects of social influence have received less systematic attention. The original Moore et al. (2024) paper [1] this study replicates was a direct response to this gap. Their findings were significant: they showed that a substantial portion of risky behaviors (20–50%) were socially triggered, and critically, that these behaviors were more likely to be shared, creating “negative chains” that actively spread insecurity. This provided the first systematic evidence that social influence is a major driver of insecure behavior, not just secure behavior. Other researchers reinforce this concern by examining the channels of negative influence [25], [26], [27]. For example, Wei et al. [27] analyzed social media platforms like TikTok and found a prevalence of anti-privacy and anti-security advice, often normalizing risky behaviors like account sharing for surveillance, highlighting the powerful role of online strangers and influencers. Akgul et al. [25] examined online advertising for VPNs on YouTube, revealing that influencers often make misleading claims about security and threats, potentially encouraging users to adopt risky configurations or forgo other security measures. Furthermore, the concept of pluralistic ignorance—where individuals publicly conform to a perceived norm they privately reject—has been recently applied to privacy by Rader [6]. This suggests users might engage in risky behaviors because they believe “everyone else is doing it,” a social expectation our study probes directly through questions about perceived societal norms. Our replication aims to test the generalizability of the Moore et al. findings and the relative importance of these various sources of negative influence in a collectivist culture.

2.4. Cultural Context

A significant limitation in the existing usable privacy and security literature is its heavy reliance on Western, Educated, Industrialized, Rich, and Democratic (WEIRD) populations [11]. The original Moore et al. study was conducted in the U.S., and the authors explicitly called for replications in different cultural contexts to understand how social triggers vary.

Cultural dimensions, such as individualism-collectivism as defined by Hofstede [11], are likely to profoundly impact social triggers. In individualistic cultures (like the U.S.), personal achievement and self-reliance are prioritized, while in collectivist cultures (like Saudi Arabia), group harmony,

loyalty, and the influence of “in-group” members (family, close friends) are paramount [8]. This difference is not merely academic; it has tangible effects on S&P behavior. Research on IT applications in the Middle East suggests that assumptions about user behavior derived from Western contexts may not hold true, emphasizing the need for culturally-sensitive design [28]. Furthermore, studies on information security in the Arab world highlight that privacy is not solely a legal entitlement but a social expectation mediated by family, faith, and institutions [29]. Dincelli [29] argues that collectivist cultures place greater emphasis on conforming to collective patterns of behavior and loyalty, which could make individuals more susceptible to social triggers, both positive and negative, originating from their in-group. Alotaibi et al. [8] further support this by noting that the offline culture of collectivism and strong social contracts in Saudi Arabia is enacted online, suggesting that familial and social ties are the primary channels for security information and, potentially, misinformation.

While a general need for increased cybersecurity awareness in Saudi Arabia has been identified [8], a systematic investigation into the negative social triggers for risky behaviors in this specific cultural context is, to our knowledge, absent. Our research directly addresses this gap by not only replicating a U.S.-based study but also by adapting the methodology to capture culturally specific factors. We aim to provide a nuanced understanding of how social dynamics unique to Saudi Arabia shape user security and privacy behaviors, particularly concerning the source and nature of negative social influence, thereby contributing to a more globally representative body of usable security knowledge.

3. Methodology

The study instrument was a structured interview protocol designed to closely mirror the core components of the original Moore et al. survey [1], while adapting the format for richer, conversational data collection. Although the interviews were conducted verbally, all responses were systematically recorded by the interviewer in a structured, open-ended survey template to support consistent coding and analysis. We explain the survey design, recruitment, participants, ethics, and limitations.

3.1. Interview Protocol Design

The protocol was organized into seven main components, each targeting a specific aspect of users’ security and privacy behaviors. Below, we outline the structure of the interview protocol and the purpose of each component.

Part-1: Instruction and Consent. At the start of each interview, participants were given a clear explanation of the study’s purpose, expected duration, and data-handling procedures. Participation was voluntary, and only individuals who provided verbal consent continued with the interview. Because the study examined potentially sensitive or risky digital behaviors, we sought to minimize social desirability

bias. Following the approach used in prior work on “privacy lies” [30], we emphasized that engaging in risky behaviors is common and not viewed negatively in this research context. Participants were reassured that we were interested in understanding real experiences rather than evaluating their choices, and they were encouraged to respond honestly and accurately.

Part-2: Usual Behavior and Context. We began by asking participants about their everyday digital habits to establish contextual baselines for interpreting their later responses. In this section, we explored how participants typically connect to the internet (e.g., reliance on cellular data versus public Wi-Fi), their familiarity with security mechanisms such as two-factor authentication (2FA), and their perceived risks associated with using public networks. We also asked them about their general perceptions of the causes of online attacks or fraud. These questions allowed us to situate participants’ later descriptions of risky behaviors and social triggers within the broader context of their routine digital practices.

Part-3: Risky Behavior Practices. Participants were asked to describe whether they had engaged in any of six common risky digital behaviors within the past six months. These behaviors were selected based on prior literature and our own assessment of practices that are broadly relevant across users and devices. The interviewer prompted participants to elaborate on the context in which each behavior occurred, allowing us to capture nuance beyond simple self-report. The six behaviors included:

- Connecting to unfamiliar or potentially unsecured public Wi-Fi networks, particularly when performing sensitive activities such as accessing financial accounts or transmitting passwords.
- Reusing identical or similar passwords across multiple online accounts.
- Downloading unofficial, pirated, or otherwise unverified software, applications, or media.
- Ignoring, postponing, or habitually delaying software or application updates.
- Sharing personal or sensitive information online with individuals they do not know, including through social media platforms.
- Sharing account credentials or allowing shared access with family members, friends, or others.

Similar to prior work reviewed in Section 2.2, we focused on behaviors that tend to occur in day-to-day digital life and that pose varying degrees of security and privacy risk depending on context. Participants were encouraged to explain why they engaged in each behavior, whether they perceived it as risky, and under what conditions they considered the behavior acceptable or justifiable. This approach enabled us to document both the practices themselves and the reasoning processes behind them. The riskiness of each behavior is described below.

- Connecting to unfamiliar or unsecured public Wi-Fi networks exposes users to the possibility of traffic interception through attacks such as man-in-the-middle interception, allowing adversaries to capture sensitive information [31]; however, this risk can be reduced when users rely on encrypted connections, VPNs, or isolated environments.
- Reusing identical or similar passwords across multiple accounts increases users' vulnerability to credential-stuffing and cross-site compromise when a single password is leaked [32], although reuse across low-value accounts with no sensitive data reduces the practical impact.
- Downloading unofficial, pirated, or unverifiable software, applications, or media may introduce malware, spyware, or other harmful code that compromises devices or personal data, though this risk is lower when users rely on reputable open-source communities or sandboxed environments.
- Ignoring or delaying software and application updates leaves known vulnerabilities unpatched, creating opportunities for exploitation by attackers [33]; however, not all updates include security fixes, and some vendors provide limited detail in release notes [34], making users unaware of the specific risks involved.
- Sharing personal or sensitive information online with unknown individuals can lead to identity theft, harassment, or stalking when details reveal a user's location or real-time activities [35], [36], though these risks are mitigated when information is shared selectively and privacy settings are properly configured.
- Sharing account credentials or granting shared access to family members, friends, or others increases the likelihood of compromise due to unsafe practices by secondary users [37], although some platforms reduce this risk through features such as temporary access links or role-based permissions.

Part-4: Behavioral Triggers. For each risky behavior that participants reported engaging in, we asked a series of questions to identify the factors that contributed to their decision. First, we asked whether they had *observed* or *heard* about someone else performing the same behavior. Participants who answered affirmatively were then asked to indicate their relationship to the person involved. The available options were friend, family member, colleague, online stranger, or media source.

We then asked whether the participant had ever been *advised* by someone to perform the behavior. If so, we again asked them to specify who provided the advice, followed by a question about whether that person had mentioned any risks associated with the behavior.

Finally, for each behavior, participants were asked to identify the *primary factor* that motivated their decision to engage in it. Based on our analysis and prior literature, we categorized these factors into four higher-level groups:

time or urgency, social or cultural norms, trust or privacy perceptions, and personal convenience. These categories allowed us to distinguish socially driven influences from individual or situational motivations.

Part-5: Sharing Practices. For each risky behavior discussed, we asked participants whether they typically shared this behavior with others. If they indicated that they did share it, we followed up by asking with whom they shared it (e.g., family, friends, colleagues, online communities, or other) and the reasons behind sharing. The reasons included motivations such as “I wanted to help them or provide a benefit” and “I just like to share.” Participants were allowed to mention multiple relationships and multiple motivations.

For participants who indicated that they did not share their behavior with anyone, we asked them to explain why. Their explanations helped us understand whether the lack of sharing was related to privacy concerns, cultural expectations, or simply perceiving the behavior as insignificant.

Part-6: Risky Behaviors of Others. Participants estimated the percentage of the public they thought engaged in each risky behavior.

Part-7: Security Attitudes and Demographics. Similar to Moore et al. [1], we incorporated the SA-6 security attitude scale [38], which offers a concise attitudinal measure more suitable for modeling risky user behaviors than intention-based instruments such as SeBIS [39]. After completing the interview portion, participants were handed a device to privately complete the SA-6 items to reduce interviewer influence and ensure more candid responses.

We then collected basic demographic information, including age, gender, education level, general IT knowledge, and nationality. The demographic section also included a simple attention-check item (i.e., one that explicitly instructs participants which option to select, rather than a deceptive trap question) to verify careful responding.

Following prior work [40], [41], we concluded by asking participants whether they had answered honestly, clarifying that indicating dishonesty would not affect their compensation or participation status.

The interviews were conducted in Arabic, the native language of the participants, to ensure maximum clarity and depth of response, and were audio-recorded with explicit consent.

3.2. Recruitment and Participants

We employed a purposive sampling strategy to recruit participants who were residents of Saudi Arabia and over the age of 18. In total, we conducted $N = 33$ structured interviews. The interviews lasted between 5 and 10 minutes. Table 1 shows the demographics of our participants.

3.3. Data Collection and Qualitative Analysis

All interviews were audio-recorded and professionally transcribed verbatim. The Arabic transcripts were then trans-

TABLE 1: Participant Demographics ($N = 33$)

Category	Group	N	%
Age	18–24	15	45.5%
	25–34	4	12.1%
	35–44	3	9.1%
	55–64	1	3.0%
Gender	Male	9	27.3%
	Female	24	72.7%
Education	High School	3	9.1%
	College	13	39.4%
	Undergraduate	13	39.4%
	Post-graduate	3	9.1%
	Other / Prefer not to say	1	3.0%
IT Knowledge	Yes*	8	24.2%
	No	25	75.8%

*For simplicity, we show the percentage of participants who chose 5 or 4 in this table.

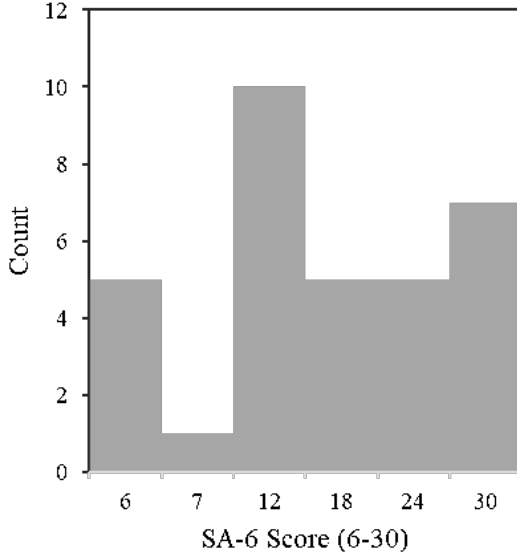


Figure 1: Distribution of SA-6 score of our participants.

lated into English by a bilingual researcher, and a second bilingual researcher reviewed a subset of the translations to ensure accuracy and preservation of cultural nuance. Full anonymized interview transcript are provided in Appendix A. We utilized Thematic Analysis to analyze the combined dataset of interview transcripts and structured documentation. The analysis followed the six-phase process outlined by Braun and Clarke [42], resulting in the identification of key themes and the quantification of related factors. The analysis process was iterative and involved multiple researchers to ensure inter-coder reliability and to mitigate individual researcher bias, particularly in the interpretation of culturally specific data.

3.4. Ethical Considerations

All participants provided informed consent prior to the interview, which included explicit permission for audio recording and the use of anonymized quotes. To mitigate the risk of social desirability bias, participants were assured of the confidentiality of their responses and were explicitly told that the researchers were interested in understanding the context, not judging the behavior. All data was anonymized immediately upon transcription, and all identifying information was removed from the final dataset.

3.5. Limitations

Measurement of behavioral triggers. Our study has limitations similar to those noted in prior work that relied on self-reported behavioral data [1], [2]. Although we asked participants to indicate whether they had observed or heard about others engaging in a behavior, whether they had received advice, and who was involved, these measures depend on participants’ recall and subjective interpretation. Self-reported triggers may not fully capture the complexity of situational and contextual factors that shape risky behaviors. We also asked participants to select the primary factor that motivated their behavior, but participants may find it difficult to isolate a single cause. Therefore, it was not possible to quantify the relative strength of the different influences.

Recruitment of participants. Our sampling strategy was purposive and focused on adults residing in Saudi Arabia. While this approach allowed us to capture culturally specific perspectives on security and privacy behaviors, it limits the generalizability of the findings beyond similar demographic groups. The sample size ($N = 33$) is appropriate for qualitative interview research, but it is not sufficient for population-level generalization. In addition, although the interviews were conducted individually and without observation, the structured setting may still have influenced how participants described their behaviors and decision-making. Future work should include larger and more demographically diverse samples to assess the extent to which these patterns hold across broader populations.

4. Results

We present the findings from the thematic analysis of the structured interviews and open-ended survey responses, organized to address our research questions: the frequency of social triggers for the risky behaviors (RQ1), the source of social triggers (RQ2), the factors of the social triggers (RQ3), and user practices in sharing the risky behaviors (RQ4). A detailed overview of all themes, including descriptions, supporting quotes, and frequencies, is provided in Table 10 of Appendix A.

4.1. RQ1: Frequency of Social Triggers

Table 2 shows the percentage of participants who engaged in each risky behavior and the frequency of behavioral triggers that led to each behavior.

Risky behaviors. The most frequently reported risky behaviors were **Password Reuse** and **Delayed Update**, with 84.8% (28/33) of our participants having reused the same or similar passwords and 63.6% (21/33) reporting having ignored or delayed software/application updates in the 6 months preceding the interview. Additionally, 45.4% (15/33) reported having shared an account with others, 33.3% (11/33) reported having downloaded illegal or unofficial content, 24.2% (8/33) reported having shared their sensitive information online, and 21.2% (7/33) reported having connected to public Wi-Fi. Overall, **100% (33/33)** of our participants reported having engaged in at least one of the six risky behaviors.

Behavioral Triggers. As shown in Table 2, social triggers were relatively common among participants who engaged in risky behaviors. On average, 65.5% of participants had observed or heard about others performing the behavior before engaging in it themselves. In contrast, fewer participants had received advice from others, with an average of 32.9% reporting that someone encouraged them to engage in the behavior. The remaining 33.3% of participants stated that they had not experienced any social trigger, which indicates that their engagement in the behavior was voluntary.

We found that the frequency of social and voluntary triggers varied across the different risky behaviors. For example, participants who engaged in password reuse, posting sensitive content, or account sharing often reported observing others performing the behavior. This pattern was even stronger for illegal downloading, where 81.8% of participants cited observation as an influencing factor. In contrast, delayed updates followed a different pattern, where advice from others (62.5%) was more common than observation.

Although some behaviors were frequently performed voluntarily, such as delayed updates and public Wi-Fi use, social influence played a significant role in several others. Nearly half of the participants who engaged in illegal downloading or account sharing reported receiving advice from peers, which shows how interpersonal communication can shape cybersecurity habits.

Most participants engaged in at least some risky behaviors on their own, but the presence of social triggers across behaviors suggests that peer influence is an important factor in risky decision-making. As discussed in Section 4.3, understanding differences in how social triggers operate across behaviors may help in designing interventions that reduce negative social influence and encourage safer cybersecurity practices.

The highest rate of social triggers was for Illegal Downloading (45.5%), followed by Delayed Update (38.1%) and Password Reuse (35.7%). Sharing Sensitive Information had the lowest rate of social triggers (12.5%), with the majority of these behaviors being driven by voluntary factors like convenience (75.0%).

4.2. RQ2: Source of Social Triggers

The original study found that participants learned about risky behaviors from a variety of social sources, including family, friends, colleagues, and online environments [1]. Consistent with their approach, we examined from whom participants in our sample had observed or heard about the risky behaviors they later engaged in. Table 3 shows the person engaging in risky behaviors that participants had observed/heard about. Across behaviors, family and friends were the most frequently reported sources. Family influence was particularly common for password reuse (71.4%) and illegal downloading (63.6%), while friends were frequently observed engaging in public Wi-Fi use (71.4%) and illegal downloading (54.5%). Colleague influence was less common overall but was notable for public Wi-Fi use, where 75.1% of participants reported observing colleagues performing this behavior. Influencers and media appeared only occasionally across behaviors, suggesting that indirect online exposure played a smaller role compared with interpersonal relationships.

We show who advised the participants to engage in risky behaviors in Table 11 of Appendix A. Please note that the number of others advising the participants was less than the number of others being observed by the participants.

To further explore how these influences differed across groups, Table 4 breaks down the sources by age, gender, and security awareness. The patterns by age were broadly consistent: both the 18–34 and 35–54 groups primarily referenced family (78.9% and 76.9%, respectively), followed by friends. Where they diverged was in exposure to online sources—participants aged 18–34 were notably more likely to report influencers or media (36.8%), whereas those aged 54 and above relied exclusively on family, friends, and colleagues. This suggests that online exposure narrows substantially with age, concentrating influence within close relationships.

Gender patterns followed similar lines. Female participants more often attributed behaviors to family (83.3%) and friends (66.6%), while male participants were comparatively more likely to cite influencers or media (44.4%). Exposure to colleagues was similar for both groups, indicating that workplace modeling may be relatively independent of gender.

Clearer differences appear when considering security-awareness levels. Participants with low SA-6 scores (6–14) showed limited influence from peers and colleagues but still frequently cited family (75%), suggesting that low-awareness individuals may rely more heavily on familiar interpersonal models. Those with medium scores (15–24) reported greater influence from friends (70.0%) alongside high family influence (80%). Participants with the highest scores (25–30) displayed a more evenly distributed pattern across family, friends, and colleagues, but notably reported no influencer or media exposure, perhaps reflecting more selective or skeptical information-processing.

TABLE 2: Frequency of behavioral triggers for risky behaviors.

Behavioral triggers		Public Wi-Fi N=7 (21.2%)	Pwd Reuse N=28 (84.8%)	Illegal DL N=11 (33.3%)	Delayed Update N=21 (63.6%)	Sensitive Post N=8 (24.2%)	Account Sharing N=15 (45.4%)	All
Social	Observation	57.1%	75.0%	81.8%	42.8%	75.0%	66.6%	65.5%
	Advice	28.5%	28.5%	45.4%	62.5%	12.5%	13.3%	32.9%
Voluntary		42.8%	28.5%	18.1%	47.6%	25.0%	33.3%	33.3%

We show the percentage of the participants who had experienced each trigger (observation of others' behavior, advice from others, or organizational enforcement) among the participants who reported having engaged in each risky behavior. Participants could select more than one trigger. Therefore, the sum of each column exceeds 100%. On the other hand, a voluntary trigger means the participant had not experienced any of the three triggers above (i.e., an exclusive option).

TABLE 3: Person engaging in risky behaviors that participants had observed/heard about.

	Public Wi-Fi (N=7)	Pwd Reuse (N=28)	Illegal DL (N=11)	Delayed Update (N=21)	Sensitive Post (N=8)	Account Sharing (N=15)	All (N=90)
Family	42.9%	71.4%	63.6%	33.3%	50.0%	40.0%	52.2%
Friend	71.4%	42.9%	54.5%	23.8%	50.0%	33.3%	41.1%
Colleague	75.1%	10.7%	18.2%	9.5%	25.0%	20.0%	17.8%
Media	0.0%	0.0%	18.2%	9.5%	25.0%	0.0%	6.7%
Influencer	0.0%	0.0%	9.1%	0.0%	25.0%	0.0%	3.3%

The first row shows the number of the participants who had observed others engaging in each risky behavior before engaging in the behavior. The sum of the percentages for each behavior exceeds 100% because the participants could select more than one type of relationship. Bold numbers highlight items greater than 50%.

Overall, these results suggest that family and friends remain the primary social sources across groups, while the range and type of influence shift with age, gender, and security awareness. In the collectivist context of Saudi Arabia, this pattern aligns with expectations that close interpersonal relationships are central in shaping security and privacy behavior.

TABLE 4: Relationships between participant demographics and those who influenced them.

	Friend	Family	Colleague	Influencer/Media
Age	18–34	63.1%	78.9%	31.5%
	35–54	61.5%	76.9%	23.0%
	≥ 54	100.0%	100.0%	0.0%
Gender	Male	55.5%	66.6%	33.3%
	Female	66.6%	83.3%	29.1%
SA-6	6–14	31.2%	75.0%	18.8%
	15–24	70.0%	80.0%	20.0%
	25–30	57.1%	57.1%	57.1%

4.3. RQ3: Factors of Social Triggers

To examine how social influences relate to participants' security practices, we conducted two regression analyses. The first assessed which factors predicted whether participants experienced a social trigger prior to engaging in a risky behavior. The second investigated whether these factors were associated with the total number of risky behaviors each participant reported. The results of both models

are presented in Table 5, and Figure 2 provides a visual summary of behavior counts across demographic groups.

The logistic regression on social triggers showed that demographic variables played only a limited role. Age and security awareness (SA-6) had coefficients near zero, indicating that neither factor meaningfully predicted the likelihood of encountering a social trigger. The coefficient for males was negative, suggesting they may have been slightly less likely to report a prior social influence compared with females, but this effect was not statistically significant. These patterns echo the findings of the original study, where demographic variables also explained little variation in whether participants encountered social triggers before engaging in risky behaviors.

In contrast, behavior type showed more substantial differences. Using software updating as the reference category, two behaviors demonstrated significant associations. Participants who reported password reuse were more likely to have experienced a social trigger beforehand, and this relationship reached statistical significance ($\beta = 1.477$, $p < 0.05$). Illegal downloading showed an even stronger association with prior social triggers ($\beta = 2.035$, $p < 0.05$), suggesting that this behavior may be strongly shaped by norms or observations within peer or family contexts. Other behaviors, such as account password reuse across platforms and posting personal information, showed positive coefficients but did not reach significance. These results suggest that while some types of risky behaviors are closely tied to social exposure, others appear to occur more independently of explicit social influence.

The ordinal logistic regression assessing the number

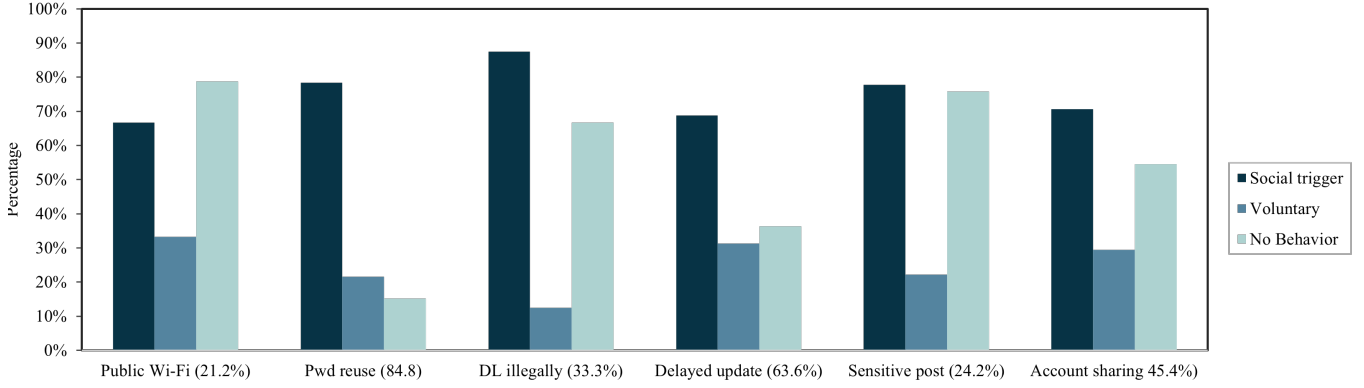


Figure 2: Percentages of public engaging in risky behavior.

of risky behaviors reported by participants showed similarly limited demographic effects. Age, gender, and security awareness all had small coefficient estimates, and none reached statistical significance. This suggests that while certain behaviors may be more socially influenced than others, the overall quantity of risky actions individuals engage in does not appear strongly related to their demographic characteristics. The column chart illustrates this pattern visually, showing similar distributions of behavior counts across age groups and genders, with only minor variation.

Together, these findings suggest that social triggers are best understood as behavior-specific rather than demographic-specific. Behaviors such as password reuse and illegal downloading appear more closely tied to social modeling, whereas the overall frequency of risky behaviors is relatively stable across participant groups. This pattern aligns with the broader observations of the original study, reinforcing the idea that social exposure plays an important role in shaping specific security practices, even when its influence is not uniform across all forms of risky behavior.

TABLE 5: Logistic regressions for risky behaviors and social triggers (coefficients and p-values).

	Social Triggers	Risky Behaviors
Age	0.016	-0.009
Male (vs. Female)	-0.428	-0.250
SA-6	0.005	-0.012
Pwd (vs. Update)	1.477*	
DL (vs. Update)	2.035*	
Account (vs. Update)	1.003	
Post (vs. Update)	1.223	

The middle column shows the results of a logistic regression explaining whether social triggers had occurred before participants engaged in risky behaviors. Of the fifteen pairwise comparisons of risky behaviors, only those pairs with a significant difference are shown in this table. The right column shows the results of an ordinal logistic regression explaining the number of risky behaviors reported by participants. Significance levels: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

In addition to the primary factors examined in the original Moore et al. study, our interview protocol incorporated two additional constructs that emerged as particularly salient in the Saudi context: *Cultural / Social Norms* and *Trust / Security Perception*. These constructs capture culturally grounded forms of risk-taking and trust-based decision-making that were not represented in the prior work but were consistently reflected in participants' explanations of their behaviors. As shown in Table 6, the distribution of primary self-reported factors varied meaningfully across the six risky behavior categories, revealing distinct behavioral patterns shaped by both cultural expectations and individual judgments of security.

Across behaviors, *Personal Convenience / Comfort* emerged as the most frequently reported factor overall (38.9%). This pattern was most pronounced for password reuse (64.3%) and illegal downloads (54.5%), where participants described prioritizing ease of use or immediate access over security considerations. Similarly, half of the participants who posted potentially sensitive content cited personal convenience or comfort, indicating that habits and preference for simplicity often override perceived risk.

Cultural / Social Norms also played a prominent role across multiple behaviors (22.2% overall), particularly for illegal downloads (45.5%), public Wi-Fi use (42.9%), and delayed updates (23.8%). Participants frequently framed these behaviors as "normal," "common," or "widely accepted," reflecting the strong influence of prevailing social practices in the region. These findings align with the collectivist cultural context, where behavior is shaped not only by personal preference but also by perceptions of what peers and family members routinely do.

Trust / Security Perception was especially influential for account sharing (66.7%), delayed updates (33.3%), and sensitive posting (25.5%). Participants often justified account sharing on the basis of interpersonal trust, particularly with close family members—an expected outcome in a culture where in-group ties are strong and deeply valued. In contrast, participants who delayed updates frequently cited a perception that updates were unnecessary, unimportant, or potentially disruptive, indicating a misalignment between users' mental models of security needs and actual technical

risks.

Finally, *Time / Hurry* accounted for 16.7% of all responses but was particularly relevant for behaviors that occur in routine or time-pressured contexts, such as public Wi-Fi use (28.6%), password reuse (25.0%), and delayed updates (23.8%). Participants frequently described needing to “finish something quickly,” “save time,” or “avoid inconvenience,” highlighting the well-documented tension between efficiency and security.

Together, these findings illustrate that risky security and privacy behaviors in Saudi Arabia arise from a combination of convenience-driven decisions, culturally shaped expectations, and trust-based assumptions, rather than purely individual motivations. The presence of strong cultural norms and interpersonal trust dynamics differentiates this context from the original U.S. study and underscores the need for culturally aware intervention strategies.

4.4. RQ4: Sharing Practices

Frequency of Sharing. Table 7 summarizes how often participants reported sharing each risky behavior and with whom these behaviors were discussed. As in the original study, sharing occurred primarily within close interpersonal circles. Family members were the most frequently cited sharing partners overall (58.3%), particularly for password reuse (81.8%) and account sharing (60.0%). Friends were the next most common group, especially for public Wi-Fi use (75.0%) and illegal downloading (71.4%). Sharing with colleagues was rare across all behaviors, appearing only in isolated cases. These patterns indicate that communication about risky behaviors is concentrated within personal rather than professional relationships.

Reasons for Sharing. Participants also provided reasons for why they shared their behaviors, shown in Table 8. The most common motivation across behaviors was a desire to help others or provide guidance based on personal experience, which was particularly prevalent for delayed updates (58.3%) and for sensitive posting and illegal downloading (both 66.6%). Desire-based sharing was present across behaviors but occurred at lower rates. Sharing prompted by someone noticing the behavior was relatively uncommon and appeared only for a small subset of participants. Overall, these results echo the original study’s findings that sharing often arises from purposeful or supportive intentions rather than casual or incidental disclosure.

Factors Associated With Sharing. To examine which factors predicted whether sharing occurred, we estimated a logistic regression model (Table 9). Demographic variables played a minimal role: age, gender, and security awareness scores showed no significant effects, consistent with the original study’s conclusion that sharing tendencies do not vary strongly across demographic groups. Behavior type, however, showed marginal trends. Participants who engaged in illegal downloading or public Wi-Fi use were somewhat more likely to report sharing compared to those who delayed updating, with effects approaching statistical significance ($p < .10$). These behaviors may therefore be more commonly discussed within peer or family contexts.

A significant positive association emerged between the occurrence of a social trigger and the likelihood of sharing ($\beta = 1.142$, $p < .05$). This suggests that behaviors influenced by social exposure may, in turn, be more likely to be communicated to others. A column chart accompanying this section illustrates the distribution of sharing across demographic groups and supports the observation

that sharing is shaped more by interpersonal dynamics than by demographic characteristics.

Overall, these results reinforce the idea that sharing is most common within close relationships and is motivated largely by the intent to help others. Behavior type plays a modest role, while the presence of social triggers significantly increases the likelihood of sharing, highlighting the interpersonal pathways through which security behaviors travel.

5. Discussion

5.1. Comparison to the Original Study

Compared to the original study by Moore et al. [1], our findings show both parallels and important cultural distinctions in how social triggers shape risky security and privacy behaviors. Similar to their results, we observed that a substantial portion of risky behaviors were influenced by observations of others or by receiving advice, reinforcing that social influence is a core mechanism behind the adoption and spread of insecure practices. However, whereas Moore et al. identified social triggers primarily through general interpersonal interactions, our study revealed that these influences are more strongly concentrated within close in-group relationships, especially Family and Friends. This difference reflects the collectivist cultural context of Saudi Arabia and suggests that social proximity may amplify the likelihood of adopting risky digital behaviors.

Both studies found that the type of risky behavior was a stronger predictor of social triggering than demographic factors. Yet the triggers in our study were shaped by additional culturally specific factors, including social norms surrounding trust, expectations of helpfulness within families, and assumptions about the commonness of certain behaviors. Moreover, participants in our study frequently justified their actions using cultural reasoning, which was uncommon in the original work. Despite these differences, the two studies converge on the conclusion that socially triggered behaviors are more likely to be shared with others, creating negative chains that propagate insecure practices. These similarities and differences highlight the importance of examining social triggers across diverse cultural contexts and suggest that interventions must account for both universal social mechanisms and local cultural dynamics.

5.2. Summary of Findings

- Risky behaviors were widespread among participants, with most reporting engagement in at least one of the six examined behaviors.
- Social triggers played a substantial role, as many participants engaged in risky behaviors after observing others or receiving advice, particularly from Family and Friends.
- Family and Friends were the dominant influencers, reflecting the strong in-group orientation of the Saudi cultural context.
- Personal factors also shaped behavior, including convenience, time pressure, and perceptions of trust or low risk.
- Sharing of risky behaviors was common, and socially triggered behaviors were more likely to be shared, contributing to negative social chains.
- Cultural and social norms influenced several behaviors, particularly those related to public Wi-Fi use and illegal downloading. Trust also played a major role, with 66.7% of participants who shared their accounts reporting that they did so because they trusted the other person.

TABLE 6: Primary self-reported factors contributing to risky behaviors across all behavior categories.

	Public Wi-Fi (n=7)	Pwd Reuse (n=28)	Illegal DL (n=11)	Delayed Update (n=21)	Sensitive Post (n=8)	Account Sharing (n=15)	All
Time / Hurry	28.6%	25.0%	0.0%	23.8%	0.0%	6.7%	16.7%
Cultural / Social Norms	42.9%	7.1%	45.5%	23.8%	25.0%	20.0%	22.2%
Personal Convenience / Comfort	28.6%	64.3%	54.5%	19.0%	50.0%	6.7%	38.9%
Trust / Security Perception	0.0%	3.6%	0.0%	33.3%	25.5%	66.7%	22.2%

Percentages represent the proportion of participants within each behavior category who selected the corresponding factor as their primary reason for engaging in that behavior. Participants selected only one primary factor. Bold numbers highlight items greater than 50%.

TABLE 7: Participants' sharing practices and person with whom they shared.

	Public Wi-Fi (N=7)	Pwd Reuse (N=28)	Illegal DL (N=11)	Delayed fUpdate (N=21)	Sensitive Post (N=8)	Account Sharing (N=15)	All
Overall shared	57.1%	39.3%	63.64%	23.81%	50.0%	33.3%	44.2%
Family	50.0%	81.8%	57.1%	40.0%	25.0%	60.0%	58.3%
Friend	75.0%	45.5%	71.4%	60.0%	50.0%	20.0%	52.8%
Colleague	0.0%	0.0%	0.0%	0.0%	0.0%	20.0%	2.8%

The first row indicates the number of participants who reported engaging in each risky behavior, and the second row indicates the percentage of participants who shared the behavior with others among those who reported engaging in the behavior. The third and subsequent rows indicate the percentage of participants who shared the risky behavior with a specific person among those who shared the behavior. The sum of the percentages for each behavior exceeds 100% because the participants could select more than one type of relationship. Bold numbers highlight items greater than 50%.

TABLE 8: Reasons for sharing risky behaviors with others.

	Public Wi-Fi (N=3)	Pwd Reuse (N=15)	Illegal DL (N=6)	Delayed Update (N=12)	Sensitive Post (N=6)	Account Sharing (N=7)	All
Desire of Sharing	33.3%	46.0%	66.6%	33.3%	33.3%	28.5%	40.8%
To Help Others	66.6%	33.3%	33.3%	58.3%	66.6%	57.1%	49.0%
Because Someone Noticed	0.0%	20.0%	0.0%	8.3%	0.0%	14.2%	10.2%

The first row indicates the number of participants who shared their risky behavior with others. The second and subsequent rows indicate the percentage of participants who shared the risky behavior with the reason for sharing the behavior. Bold numbers highlight items greater than 50%.

- Misconceptions about how common risky behaviors are appeared to reinforce further risky actions, as some participants assumed others behaved similarly. Overall, the findings suggest that both social influence and individual decision-making factors interact, shaped by Saudi-specific cultural norms, to sustain and spread risky security and privacy behaviors.

5.3. Recommendations

Our findings show that risky behaviors in the Saudi context are shaped by a combination of social triggers, cultural norms, perceived convenience, and trust in familiar individuals. Similar to the original study [1], these influences suggest that effective interventions must account for the social and cultural mechanisms through which risky practices spread. Based on our results, we outline several recommendations for reducing the impact of negative social triggers on security and privacy behavior.

Leveraging positive social influence while reducing negative effects. While negative social chains can promote risky behavior,

positive chains can also encourage secure practices. Prior work, including Das et al. [2] and the studies reviewed in Section 2.3, showed that advice from trusted individuals often leads to safer behaviors. Programs should therefore aim to strengthen supportive peer influence by equipping users with simple and transferable security practices that can spread through the same social pathways that currently facilitate risky ones. In our findings, participants frequently engaged in risky behaviors after observing or receiving advice from Family and Friends. This indicates that interventions should prioritize small social groups rather than individuals. Community-based security campaigns that highlight secure alternatives may shift group norms toward safer practices. Because social triggers can also promote secure behavior, interventions should reinforce these positive pathways by providing trusted in-group members with accurate and practical guidance.

Behavior-specific awareness rather than demographic targeting. Consistent with the original work [1], the type of behavior had a stronger influence on social triggering than demographic factors. Behaviors such as password reuse, delayed updates, and illegal downloading showed the highest levels of social and cultural

TABLE 9: Logistic regression for social triggers.

	Coeff	p-value	
Age	0.011	0.127	
Male (vs. Female)	-1.180	0.659	
SA-6	-0.042	0.284	
DL (vs. Update)	1.675	0.054	
Wi-Fi (vs. Update)	1.920	0.097	
DL (vs. Account)	1.354	0.142	
Wi-Fi (vs. Account)	1.721	0.147	
DL (vs. Pwd)	1.124	0.149	
Wi-Fi (vs. Pwd)	1.249	0.192	
Social Trigger	1.142	0.024	*

Of the fifteen pairwise comparisons of behaviors, only the most significant differences are shown. Significance levels: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

influence. Awareness efforts should therefore focus on specific behaviors that are most prone to negative social chains. Clear explanations of risk, paired with concrete examples of potential harm, may help counteract convenience-driven decisions.

Addressing cultural and social norms. A substantial portion of participants justified risky behaviors through cultural or social expectations, including assumptions that such practices are normal within their families or communities. Interventions should explicitly address these norms by highlighting how common secure practices actually are, thereby correcting misperceptions. Community-anchored messaging, where respected local or familial groups model secure behavior, may help reshape expectations.

Improving perceptions of security and trust. Trust in familiar individuals played a major role, particularly in account sharing and decisions involving device or network use. Educational materials should clarify that trust in a person does not eliminate technical risk, and that secure alternatives such as temporary access mechanisms or privacy-preserving sharing options exist. This may help reduce the assumption that harm is unlikely when the other party is well known.

Countering overestimation of how common risky behaviors are. Many participants believed that certain risky behaviors were widely practiced, which may have reinforced their own decisions. Interventions can reduce this bias by presenting factual information about how frequently informed users avoid such behaviors. Prior research shows that correcting misperceptions can reduce risky conduct in other domains, suggesting similar strategies may be effective for S&P behaviors.

Enhancing practical security support. Because convenience was a primary factor for several behaviors, tools that simplify secure actions may reduce risky choices. Examples include easier update mechanisms, clearer notification prompts, and simplified password management techniques. When secure options are perceived as less burdensome, social triggers that motivate risky alternatives may lose their influence.

6. Conclusion and Future Work

This replication study successfully investigated the negative effects of social triggers on user security and privacy behaviors

in the unique cultural context of Saudi Arabia. We confirmed the prevalence of social triggers and, most importantly, identified a significant cultural difference: the overwhelming dominance of family and friends as the source of these triggers. This finding underscores the necessity of conducting security research in diverse cultural contexts and provides a strong basis for developing culturally-aware security interventions.

Future work should explore the specific mechanisms of the "Trust Factor" in more detail, perhaps through a dedicated qualitative study. Furthermore, a quantitative study with a larger, more representative sample would be beneficial to statistically generalize the findings and explore correlations between demographics and the type of social trigger.

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Appendix

A Interview Transcript

A Study on Online Behavior

Interviewer (I): Welcome, and thank you for your time. This study explores the negative effects of social triggers on user security and privacy behaviors, specifically how social influence or advice from others may lead individuals to engage in risky online actions. Participation will take about 10–12 minutes. It is completely voluntary and anonymous. No personal information will be collected, and all responses will remain confidential.

We are interested in this topic as researchers, so we kindly ask that you provide honest and accurate responses. Your input is vital and will help us gain meaningful insights into this research area. Before we begin, I need to confirm:

- You are over 18 years old.
- You agree to participate in this study.
- You agree to have your voice recorded for research purposes.

P: Yes, I agree.

Part 1: General Digital Behavior

I: When you are outside the house, such as in a café, airport, or other public place, how do you usually connect to the internet — your private mobile data or public Wi-Fi?

P: Private.

I: Do you know what two-factor authentication is, and do you use it?

P: No.

I: For example, when you receive a verification code by SMS and enter it after typing your password. Do you recognize that, and do you use it when available?

P: Oh, yes. I do when the application has that option.

I: When you use public Wi-Fi, do you expect any risks to your data?

P: Of course.

I: What kinds of risks do you think could occur?

P: Possible data breaches or access to information on my phone.

I: In your opinion, do people in Saudi Arabia generally care about these risks?

P: No.

I: Have you or someone you know ever been a victim of cyber fraud?

P: Yes, one of my social media accounts was hacked and deleted.

I: What do you think caused it?

P: Messages asking for information or ID numbers.

I: Why do you think people fall for these attempts?

P: Because they do not have the knowledge to recognize the risks.

I: Now we will ask about behaviors that are very common among many people. There are no right or wrong answers, we only want to understand how people usually behave. During the past 6 months, have you done any of the following behaviors?

Part 2: Risky Behaviors, Triggers and Sharing Practices

Public Wi-Fi

I: Have you used public Wi-Fi and entered important information, such as card details or verification codes?

P: Yes.

I: Before doing this, did you see someone else doing it or hear anyone talk about it?

P: Yes.

I: Who did you notice — family, friends, famous people?

P: Family and friends.

I: Did anyone advise you to do this?

P: No.

I: What influenced you — advice, observation, personal desire, or urgency?

P: Observation, and it is something common in our culture.

I: After doing this behavior, did you tell anyone about it?

P: Yes.

I: Who did you share it with?

P: Family and friends.

I: Why did you talk about it?

P: I like to share information — either to warn others or just to share.

Password Reuse

I: Do you reuse the same password for more than one account?

P: Yes.

I: Before doing this, did you see someone else doing it?

P: Many people, yes.

I: Who did you notice?

P: Family, friends, people in general.

I: Did anyone advise you to do this?

P: No.

I: What influenced you the most?

P: Lack of time, cultural habits, and urgency.

I: Did you tell anyone that you reuse your passwords?

P: No.

I: Why not?

P: To protect my privacy.

Illegal Download

I: Have you downloaded programs, movies, or songs from unofficial websites?

P: No.

I: Or watched them on unofficial sites?

P: I have never watched.

Delayed Updates

I: Do you usually ignore or delay updates?

P: Automatic updates are enabled, so I am not normally notified.

Sharing Sensitive Information Online

I: Have you shared personal information on social media with people you do not know?

P: No.

Account Sharing

I: Do you share your accounts or secret numbers with family or friends?

P: Maybe yes.

I: Did you see someone else doing this?

P: Yes.

I: Who?

P: Friends.

I: Did anyone advise you to do this?

P: No, it was something I decided myself.

I: What influenced you?

P: It is something habitual, not necessarily cultural.

I: Did you tell anyone about this behavior?

P: No.

I: Why not?

P: There is no need — it is a normal thing to do.

I: If you did tell someone, why would you?

P: Maybe to help.

Part 3: Perceptions of Society and Technology

I: In your opinion, what percentage of people engage in these behaviors (password reuse, account sharing, unofficial downloads)?

P: 90%.

I: Do you think society sees these behaviors as dangerous or normal?

P: Normal.

I: Do you feel that culture encourages people to be cautious about digital security?

P: Yes.

I: Where do you usually get your technical information — friends, family, social media, school, or university?

P: University and family.

I: Do you usually follow experts' advice or ignore it?

P: I listen to it.

I: How would you rate your cybersecurity knowledge on a scale from 1 to 5?

P: Beginner — maybe 2 or 3.

Part 4: Demographics

I: I will give you the device. Please take a moment to answer the six SA-6 questions on your own, and then fill out the short demographic section that follows on the same form.

P: (Participant completes the SA-6 items and the demographic questions on the device.)

I: Thank you for your participation and for your time today.

P: Thank you.

B Codebook with Frequencies

TABLE 10: Thematic Codes, Descriptions, Example Quotes, and Frequencies.

Code	Description	Quote	Freq.
Influencing Factor: Time / Hurry	Risky actions performed because participants were in a hurry, prioritizing speed over security.	"I was in a hurry, so I did it without thinking about the risks."	12
Influencing Factor: Cultural / Social Norms	Risky behaviors driven by cultural expectations or common practices.	"It's common in our family to know each other's account details. Everyone does it."	17
Influencing Factor: Personal Convenience / Comfort	Behaviors motivated by convenience, simplicity, or comfort over security.	"Using one password is easier for me — I don't want to remember a bunch of different ones."	28
Influencing Factor: Trust / Security Perception	Behaviors influenced by trust in close relationships, reducing perceived risk.	"I'm not worried. I trust my sister with my account; she'd never misuse it."	7
Motivation for Sharing: Because Someone Noticed	Participants shared only when someone else observed the behavior.	"Someone saw a post I shared online about an activity I participated in."	5
Motivation for Sharing: To Help Others	Prosocial sharing intended to help others.	"I only talked about it because the update was making my phone lag."	22
Motivation for Sharing: Desire to Share	Sharing motivated by wanting to participate in conversations.	"I always like to participate in conversations."	22
Reason for Not Sharing: Opportunity / Chance	Participants did not have time or opportunity to share.	"I didn't have time or opportunity to tell anyone."	18
Reason for Not Sharing: Not Worth Mentioning / Common Practice	Participants felt the behavior was too common to mention.	"I don't feel it was worth mentioning; it's common between people."	34
Reason for Not Sharing: Privacy / Personal Preference	Participants preferred to keep digital habits private.	"What I do on my accounts is personal, so I don't like sharing."	13
Perceived Risk (Wi-Fi): Data Theft / Hacking	Concern about stolen data or hacked accounts.	"Maybe they could steal personal data."	15
Perceived Risk (Wi-Fi): No Risk Perception	Participants perceived no risk in public Wi-Fi.	"No, I don't think there is any risk."	8
Perceived Risk (Wi-Fi): Accessing Financial Information	Fear of attackers accessing bank accounts.	"Maybe accessing my financial accounts to steal money."	5
Perceived Risk (Wi-Fi): General Risk Awareness	General acknowledgment of risk.	"I'm aware that public Wi-Fi isn't completely safe."	2
Perceived Risk (Wi-Fi): Device Control / Malware	Fear of remote control or device compromise.	"Maybe they can access my device and control it."	1
Perceived Risk (Wi-Fi): Intercepting Communications	Concern that attackers could monitor activity.	"Someone could observe my online activity."	2
Perceived Causes of Online Fraud: Risky Behavior / Carelessness	Fraud attributed to user negligence.	"People fall for fraud because they aren't careful with links."	4
Perceived Causes of Online Fraud: No Incident	Participants had not witnessed fraud.	"Nothing happened to me or anyone I know."	13
Perceived Causes of Online Fraud: Lack of Knowledge	Fraud attributed to low digital literacy.	"People fall for fraud because they don't really know much about it."	9
Perceived Causes of Online Fraud: Technical / Systemic Causes	Fraud linked to systemic or technical issues.	"We didn't have specialists to help us."	3
Perceived Causes of Online Fraud: Temptations / Emotional Factors	Fraud attributed to emotional manipulation.	"People fall for scams because the offers seem tempting."	4

C Detailed Results

TABLE 11: Person who advised participants to engage in risky behaviors.

Relationship	%
Friend	30.3%
Family	57.6%
Media(e.g.,social media, TV programs)	18.2%
Influencer	3.0%

Table 11 indicates from whom the participants were advised to engage in risky behaviors. Participants were primarily influenced by family, friends, media, and influencers similar to Table 3 (i.e., observation of others).